Lesson 23: Quasi-Experimental Research

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 8.1: Prelude to Quasi-Experimental Research; 8.2: One-Group Designs; 8.3: Non-Equivalent Groups Designs; 8.4: Quasi-Experimental Research (Summary).

In this lesson

You will learn what quasi-experiments are and why their results are harder to interpret than those of true experiments: several alternative explanations can look like a treatment effect. In English, you will practise proposing alternative explanations (*might have caused*) and comparing designs (*is like … in that*).

Learning goals

After this lesson you should be able to:

- Define quasi-experimental research and explain why its internal validity lies between that of non-experimental studies and true experiments.
- Describe the three one-group designs and the five nonequivalent groups designs.
- Identify threats to internal validity and explain how a control group or a switching replication reduces them.
- Propose alternative explanations with *might / may / could have* + past participle, and compare designs with *is like / is unlike … in that*.

Key language

Term or phrase	Meaning	Example
quasi-experimental research	Research that manipulates an independent variable but lacks random assignment to conditions or orders of conditions; generally used in field settings to test a treatment.	An anti-drug program in one school
pretest-posttest design	The dependent variable is measured (pretest), a treatment is given, and it is measured again (posttest).	Attitudes to drugs before and after the program
interrupted time-series design	A series of measurements of the dependent variable taken both before and after a treatment.	Weekly productivity before and after shorter shifts
nonequivalent groups design	A between-subjects design without random assignment, usually because participants are in preexisting groups.	Two third-grade classes

Term or phrase	Meaning	Example
switching replication	The control group receives the treatment later, so the effect should appear in each group at a different time.	Patients exercise first, students later
history	Other events between pretest and posttest that could cause the change.	A celebrity dies of a drug overdose
maturation	Developmental changes in participants between measurements that offer an alternative explanation.	Children become less impulsive
testing	The pretest itself affects responses at posttest.	The attitude questionnaire starts conversations about drugs
instrumentation	The measuring instrument, including human observers, changes over time.	Bored participants answer carelessly at posttest
regression to the mean	Someone with an extreme score on one occasion tends to score less extremely on the next.	A bowler who averages 150 bowls 220, then lower
spontaneous remission	Improvement in a psychological or medical problem over time without any treatment.	A cold improves within a week anyway
one often cannot conclude this with a high degree of certainty	Limits a causal claim	"...because there may be other explanations"
one must generally be very cautious about inferring causality from	Warns about a design	"...from pretest-posttest designs"

Rules and explanations

The ideas

What is a quasi-experiment? *Quasi* means "resembling". The independent variable is manipulated, but "either a control group is missing or participants are not randomly assigned to conditions" (or the order of conditions is not counterbalanced). Manipulation before measurement removes the **directionality problem**; without random assignment, **confounding variables** remain. So in internal validity, quasi-experiments are "generally somewhere between non-experimental studies and true experiments".

One-group designs (within-subjects in nature)

Design	What happens	Evaluation
Posttest only	Treatment, then one measurement (the Brand X claim: "80% of women noticed their skin looked bright")	The weakest: with no comparison, the result "means little to nothing"
Pretest-posttest	Measure, treat, measure again (the anti-drug program)	Order cannot be counterbalanced; many alternative explanations
Interrupted time series	Many measurements before and after (factory shifts cut from 10 to 8 hours)	Shows whether a change is more than normal variation

In the hypothetical attendance example, one measurement at Week 7 and one at Week 8 could make an ineffective treatment look effective; the full series shows a drop that is "nothing more than normal week-to-week variation".

Threats to internal validity. History, maturation, testing, instrumentation, regression to the mean and spontaneous remission (Key language) are alternative explanations for a pretest-posttest change. Regression to the mean is a problem when participants are selected *because* of extreme scores. A control group faces the same threats, so it reveals the actual treatment effect (if any), but then the design is no longer one-group.

Does psychotherapy work? Eysenck (1952) summarised 24 pretest-posttest studies: about two thirds of patients improved. But archival records showed similar recovery rates *without* psychotherapy, so the improvement might be spontaneous remission. He did not conclude that psychotherapy was ineffective, only that there was no evidence that it was effective. Later randomized experiments, summarised by Smith, Glass and Miller (1980), found that about 80% of treated participants improved more than the average control participant. In studies of depression treatments, waitlist controls improved by 10 to 15% on average before any treatment (Posternak & Miller, 2001).

Nonequivalent groups designs (between-subjects in nature)

Design	Structure and book example
Posttest only nonequivalent groups	One class is taught fractions a new way, another class is not; compare at the end
Pretest-posttest nonequivalent groups	Both schools are tested before and after; only one gets the program. Does it improve *more*?
Interrupted time series with nonequivalent groups	Add a company that does not change its shift length
Pretest-posttest with switching replication	Exercise for depression: patients start, students start later; depression measured three times
Switching replication with treatment removal	Patients stop exercising when students start

Choosing similar groups removes some confounds, but others may remain, including differential history (asbestos closing one school for a month). A switching replication has a built-in replication and

better control of history, maturation and instrumentation: an outside event is unlikely to coincide with *both* introductions of the treatment. These designs are the highest in internal validity. With random assignment, a pretest-posttest nonequivalent groups design becomes a true experiment.

Language focus

1. Alternative explanations with modal verbs. The book presents them as possibilities, not facts:

- "Other things *might have happened* between the pretest and the posttest that caused a change from pretest to posttest."
- "it *might have been caused* by the difference between the teaching methods—but it *might have been caused* by any of these confounding variables."

Pattern: *might / may / could* + *have* + *past participle* for a possible past event (passive: *might have been caused*). For a general possibility: "the treatment *might be* responsible for the improvement."

2. Comparing designs point by point.

- "the interrupted time-series design is like a pretest-posttest design in that it includes measurements of the dependent variable both before and after the treatment. It is unlike the pretest-posttest design, however, in that it includes multiple pretest and posttest measurements."

In that means "in the respect that" and needs a full clause.

> **Note for Arabic speakers** - Arabic expresses a past possibility with a particle and a past-tense verb, so learners often write *might caused* or *might has caused*. English needs modal + *have* + past participle: *might have caused*.

Practice

1. Reading questions

1. Why does quasi-experimental research eliminate the directionality problem but not the problem of confounding variables?
2. Why does the Brand X statistic mean "little to nothing"?
3. Why is the order of conditions in a one-group pretest-posttest design not counterbalanced?
4. If only students with extremely favourable attitudes towards drugs receive the program, why will their posttest scores probably be lower even if it has no effect?
5. What did Eysenck conclude, and what did he *not* conclude?
6. In a pretest-posttest nonequivalent groups design, how can history or maturation be separated from a treatment effect? What threat remains?
7. In the switching replication example, why would a weather change make depression decrease in both groups?
8. What extra evidence does treatment removal provide?

2. From the book (8.4 Exercises)

1. Practice: "Imagine that two professors decide to test the effect of giving daily quizzes on student performance in a statistics course. They decide that Professor A will give quizzes but Professor B will not. They will then compare the performance of students in their two sections on a common final exam. List five other variables that might differ between the two sections that could affect the results." Also name the design.
2. Discussion: "Imagine that a group of obese children is recruited for a study in which their weight is measured, then they participate for 3 months in a program that encourages them to be more active, and finally their weight is measured again. Explain how each of the following might affect the results: regression to the mean, spontaneous remission, history, maturation." Then suggest one improvement from Section 8.3.

3. Language A. Rewrite each statement as an alternative explanation with *might / may / could have* + past participle.

1. An anti-drug program aired on television.
2. The participants became bored with the measure at posttest.
3. Asbestos was found in one of the schools.

B. Compare in two sentences with *is like … in that* and *is unlike …, however, in that*:

4. the switching replication with treatment removal design and the pretest-posttest design with switching replication.

Speak with your tutor

- Explain how a quasi-experiment differs from a true experiment and from a correlational study. Ask your tutor to correct the content and your comparison language.
- Describe an advertisement based only on "after" data, like the Brand X claim, and say which comparison is missing.
- In PSYC 110 Lecture 20 (The Good Life: Happiness), Professor Bloom argues that getting better after therapy "doesn't tell you anything" because of regression to the mean, and describes a study in which depressed people were randomly given therapy or put on a waiting list. Compare his argument with Eysenck's: which threat does each emphasise, and why is Bloom's example a true experiment?
- Present your quasi-experimental design (Writing task) and ask your tutor to attack its internal validity. Defend it.

Writing task

Your project: a quasi-experimental option. Write 200-250 words describing a quasi-experimental version of your study of sleep and memory in university students. For example, a sleep-education workshop could be offered in one residence hall, with a similar hall as a nonequivalent control group and a memory test before and after; or memory scores could be measured weekly before and after the university moves its first lecture from 8:00 to 10:00 (an interrupted time series). If you work on your own topic, design a quasi-experiment that tests a treatment or a real change related to it.

Include: the book's name for your design; why random assignment is not possible; the treatment, groups (or measurement times) and dependent variable; three threats to internal validity, written with *might / could have* + past participle; one sentence comparing your design with a true experiment.

Self-editing checklist:

- I used the exact design name from Chapter 8.
- I explained why participants cannot be randomly assigned.
- I named three threats and how my design reduces each.
- My modal verbs are correct: *might have caused*, not *might caused*.
- My comparison uses *in that* + a full clause.
- I limited my causal claim.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Eysenck's 1952 article, "The Effects of Psychotherapy: An Evaluation" (full text, linked in Section 8.2): https://psychclassics.yorku.ca/Eysenck/psychotherapy.htm
- PSYC 110 Lecture 20, The Good Life: Happiness (video; the first part asks whether therapy works): https://oyc.yale.edu/psychology/psyc-110/lecture-20

Answer key

1. Reading questions

1. The independent variable is manipulated before the dependent variable is measured, so the direction is clear; but without random assignment or counterbalancing, other differences between conditions are likely (8.1).
2. There is no comparison group: we do not know how many women would have noticed bright skin without the cleanser (8.2).
3. It is typically not possible to test someone in the treatment condition first and then in an "untreated" control condition (8.2).
4. Regression to the mean: people selected for extreme scores tend to score less extremely next time (8.2).
5. Similar patients recovered at about the same rate without psychotherapy, so the improvement might be spontaneous remission. He did not conclude that psychotherapy was ineffective; he called for properly planned experiments (8.2).
6. Compare the change in both groups: a treatment effect makes the treated group change more, while history or maturation should change both groups similarly. An event at only one school remains a threat (8.3).
7. A weather change would affect both groups at the same time, but the treatment starts at different times, so only a real treatment effect follows the staggered starts (8.3).
8. A reversal when the treatment is removed, replication in two groups staggered over time, and evidence on whether the treatment still works after it is withdrawn (8.3).

2. From the book

1. (example answer) A posttest only nonequivalent groups design (no random assignment to sections). Possible variables: the professors' teaching style; the time of day; students' prior maths ability; students' motivation (motivated students may choose one professor); class size; the classroom; the amount of homework. Any five plausible differences are correct.
2. (example answer)

- *Regression to the mean*: the children were selected for an extreme weight, so their second measurement will tend to be less extreme, which could look like a program effect.
- *Spontaneous remission*: some children's weight problems might improve over 3 months without any program.
- *History*: new school meals, a health campaign on television or a season with more outdoor play could change weight.
- *Maturation*: children grow, and growth changes weight whatever the program does, so it could hide or exaggerate an effect.
- *Improvement*: measure a nonequivalent control group of obese children at the same times, or start the program in that group 3 months later (switching replication).

3. Language

1. An anti-drug program might have aired on television.
2. The participants may have become bored with the measure at posttest.
3. Asbestos could have been found in one of the schools.
4. (example answer) The switching replication with treatment removal design is like the pretest-posttest design with switching replication in that both groups receive the treatment at different times. It is unlike it, however, in that the treatment is removed from the first group when it is added to the second.

My notes